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INGENIERÍA Y MATERIALES ®

TECHNICAL CATALOG · ANCHORING

Resin cartridge for rock anchoring

Polyester resin fastening system to install anchor bolts and bars in rock and concrete: a secure, fast and lasting bond.

MINING · TUNNELS · CIVIL WORKS

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01 — DESCRIPTION

What is the resin cartridge?

The resin rock anchor cartridge is a fastening system used to install anchors in rock or other high-strength materials.

It uses a special resin as the anchoring medium, giving a secure, lasting hold in rock of many compositions and hardnesses. Its structure, how it works and its applications are described below.

Application

It is used mainly in construction, mining and rock drilling. Installation is simple: an injection gun mixes the resin components as they are injected into the hole drilled in the rock.



APPLICATIONnstalling anchors at the excavation face.

02 — BENEFITS

Advantages and common applications

Secure, lasting hold

The resin forms a very strong bond with the rock and keeps anchors from shifting or loosening over time.

Less damage to the rock

Because no invasive mechanical fixing is needed, damage to the rock structure is minimized.

Adaptability

Works in a wide range of rock types and applications, from light work to large projects.

Common applications

- Fixing reinforcing bars in tunnel or shaft construction.
- Installing anchors for support structures in underground mining.
- Mounting steel or concrete structures on rock.
- Anchoring systems for heavy equipment in rocky ground.



SITE Anchors and structures fixed on site.

03 — DATA SHEET

Purpose and technical data

Purpose

Polyester resin product formulated to anchor bolts, anchor bars and load-bearing fixings, mainly in rock and concrete.

Technical data

Direct tensile pull-out strength in rock, with a 50 mm diameter hole and a 25 mm diameter ribbed steel anchor bolt. Anchored 15 cm, after 24 hours at 23 °C:

114.4 kN	**7.9** MPa	**1.8** kg/dm³
Maximum load	Tensile strength	Density at 20 °C

Tests: Pontificia Universidad Católica, report REP 23602.

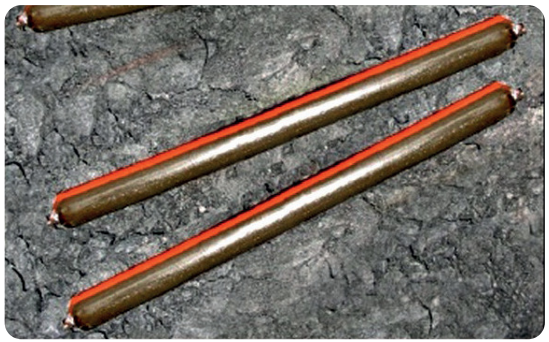


PRODUCT Resin cartridges in their different sizes.

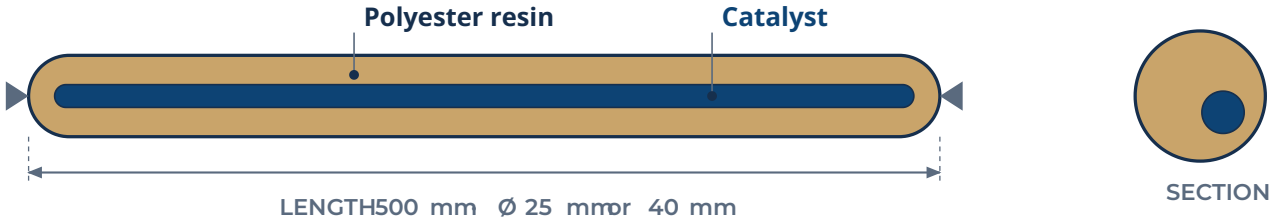
04 — PROPERTIES

Properties

- Strong adhesion to rock, concrete and steel.
- Very fast placement and setting.
- High anchoring capacity in a short time.
- Thixotropic material.
- High mechanical and chemical resistance.
- Once cured, the resin resists acids and alkalis between pH 2 and 10.
- Does not weaken even when submerged in water.
- Provides permanent anchorage.
- Reduces labor costs.
- Fully cured, it is stronger than the rock and most materials it is used in.
- Resists heat distortion up to 100 °C.
- Installation is not affected and strength does not drop as long as the resin and its wrapper are at 10 °C when installed.
- The gel time (plastic state before hardening) is given for each type and can range from 1 to 15 minutes. Initial and final set are not distinguished because the interval between them is very short.



Cartridges ready to insert in the hole.



STRUCTURETwo-component cartridge: resin and catalyst.

05 — PACKAGING AND STORAGE

Usage recommendations

Packaging

- 25 mm × 500 mm cartridge
- 40 mm × 500 mm cartridge

Consumption

Varies with the hole diameter and length and the anchor bar length.

Storage

As stated on the packaging, provided they are kept in cool, covered places in their original, tightly closed packaging.

Do not expose the cartridges to sunlight.

Cleaning

When the job is finished, clean tools with an acetone-type solvent.



CARTRIDGES Resin cartridges for rock anchoring.

06 — SAFETY

Precautions

When handling any chemical product, avoid direct contact with eyes, skin and airways.

- Wear rubber gloves and safety glasses.
- In case of eye contact, rinse immediately with plenty of water, holding the eyelids open.
- Wash your hands before eating.







07 — STEP-BY-STEP INSTALLATION

How to use

01 Drill to a depth equal to the bolt length, so the threaded part stays outside the hole. The bolt tip should be cut at a bevel.

02 Clean the hole as well as possible, preferably with compressed air.

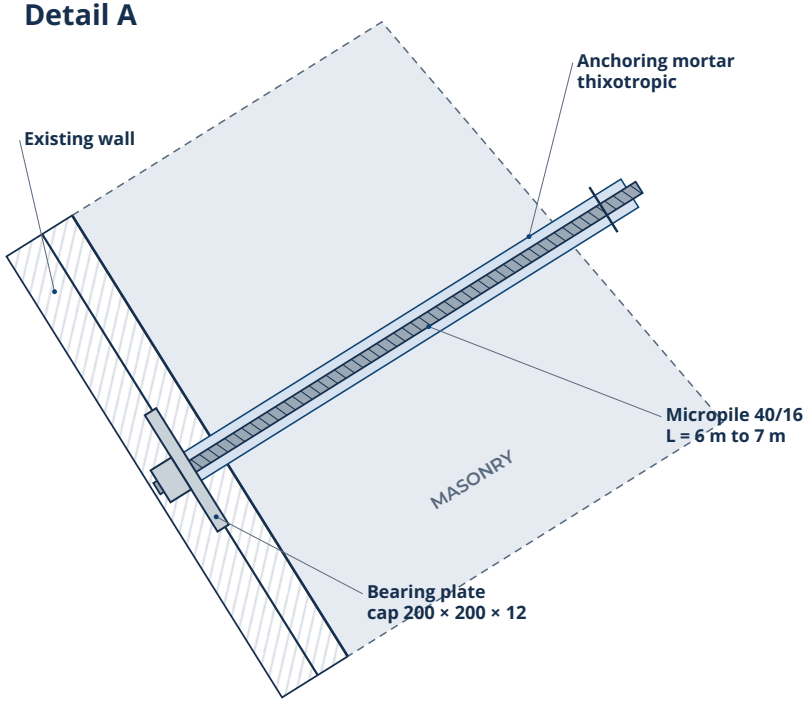
03 Insert the cartridges into the hole, allowing 10 to 20 % extra resin to fill irregularities or cracks.

04 Insert the bolt while rotating it mechanically through the cartridges to break them and mix the resin with the catalyst. You can use the same drilling rig.

05 When the bolt reaches the bottom, keep rotating for 90 to 100 seconds to ensure the components are well mixed.







DETAIL A Micropile installation with anchoring mortar.

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Contact us

Technical advice, quotes and custom fabrication for your next project.

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